W3 Lab B: Sound Design and Dynamic Audio: Quick Card

A two-page summary. Every step, with pictures, is in the **Worksheet** (on Blackboard and in the project's `Docs` folder). If in doubt, follow the Worksheet.

Today

Rebuild the **mystery sound** two ways: **Part A, Recorded** (Campfire A) and **Part B, Procedural** (Campfire B). Hear it again: `Assets/W3B_Fire/Audio/Mystery`.

	First half	Second half
Groups 1, 2	**Part A**	Part B
Groups 3, 4	**Part B**	Part A

I will call the swap. Then each group plays its fire to the room.

Ethics and safety

- **Ask** before recording anyone. Record in a **corner of the lab**.
- **Never leave the recorder alone. Format the SD card** before you hand it back.
- Computer volume **low**. Part B: **Gain** starts at 0.1. Headphones **off** for the first Play after a code change.

Get the project (GitHub Desktop)

1. **github.com/RichSF1210/W3B_FireLab**: green **Code** button, **Open with GitHub Desktop**. Local Path: a folder **not** synced to OneDrive or iCloud (Windows `C:\Unity\`). **Clone**. Never Commit or Push.
2. Own computer: open **Unity Hub**, sign in, free **Personal** licence.
3. **Windows:** double-click `Assets/W3B_Fire/Scenes/W3B_Student_Shell`. **Mac:** Unity Hub, **Add > Add project from disk**, the `W3B_FireLab` folder. Open **W3B_Student_Shell**.

First: listen

How many **parts**? Each one: high or low? **Constant** (one **loop**) or **short** (**four or more** short files)? Name them.

Part	Constant or short?	Part A files	Part B block

One campfire at a time: click the other campfire in the Hierarchy, **untick** the box next to its name (stop Play first). **One layer at a time:** tick **Mute** on its Audio Source.

Part A: Recorded (Campfire A)

1. **Record** with a Zoom in a corner. Loudest moments about **-12 dB**, **never red** (clipping cannot be fixed). Several takes per part.
2. **Files off** at the shared card reader. Then **format the SD card**.
3. **Audacity:** rename files `name_01`, `name_02` ... Drag them all in. **Solo** one track.
   - **Constant part:** loop button on; drag 3 to 5 steady seconds on the ruler; right-click the ruler, tick *Setting a loop region also makes an audio selection*; check **blue over the waveform**.
   - **Short part:** select one sound from its very start to where it dies away. Repeat: at least four files.
   - **File > Export Audio:** WAV, **Mono**, 48000 Hz, 24-bit, Export Range **Current Selection**.
4. **Into Unity:** drag your WAVs into `Audio/My_Recordings`. Stereo? Tick **Force To Mono**, Apply. No recordings? Use `Audio/Fire_Recorded`.
5. **Constant part:** layer, **Add Component > SampleLoopLayer**, loop onto the Audio Source's **Audio Generator**.
   Short part: layer, **Add Component > MyRandomPlayer**, lock the Inspector (padlock), select all short files, drag onto the word **Clips**, unlock.

Part A: your one line

`Scripts/Student/MyRandomPlayer.cs`, search **pitch**. Predict, then delete the two slashes:

```
source.pitch = Random.Range(0.85f, 1.2f);   // a random pitch every play
```

Save, go back to Unity, **wait for it to compile the C# code**, press Play. The files never change; the playback never repeats.

Part B: Procedural (Campfire B)

1. **One script per layer.** Layer 1: **Add Component > NoiseLayer** (a constant part). Layer 2: **Add Component > BurstLayer** (a short part). Layer 3: either, for a third part.
2. **One at a time:** Play, move the sliders, **describe what you hear**, write the values down (they reset on Stop). Then **read the comment** in the script: was your ear right?
3. **Resonance** works only with **Band Pass**.

Step	NoiseLayer	BurstLayer
Source	White noise	White noise
Filter	Cutoff: low = roar, high = hiss	Brightness: high = small, hard things
Envelope	A slow wobble	Instant start, quick fade
Randomness	Random wobble	Random timing and size

Part B: your one line (RTPC)

Campfire B: **Add Component > MyDynamicLink**. **Before Play**, fill **Fire** (Campfire B), **Burst Layer** and **Noise Layer** (drag from the Hierarchy). In `MyDynamicLink.cs`, pick **ONE** line, A or B, and delete its two slashes:

```
// A) more snaps as the fire grows
if (burstLayer) burstLayer.burstsPerSecond = Mathf.Lerp(1f, 40f, i);
// B) a brighter fire as it grows
if (noiseLayer) noiseLayer.cutoff = Mathf.Lerp(200f, 2000f, i);
```

Save, compile, Play. Select Campfire B and drag **Intensity**: is it more than volume? Then try the other line. **RTPC** (Real-Time Parameter Control): a value from the game drives a parameter of the sound.

Read the code: what would `Mathf.Lerp(40f, 1f, i)` do? Change one number and listen.

Stuck?

No sound	Campfire ticked? Layer muted? Gain above 0?
Red error	Typo in your line: check the `;`. Double-click the error.
NullReference	Empty slot. Stop, fill it, **Clear** the Console.
Changes vanished	Made while playing. Redo when stopped.
Only one script plays	Two on one layer: one per layer.
Mac: "damaged"	**Cancel**, never Move to Bin. Use GitHub Desktop.
Too loud	Stop. Volume and Gain down.

Stretch

- **Combine** recorded and procedural on one fire.
- **Compare** with `W3B_Complete_Reference`.
- **Interact:** `AddLogOnKey`, press **L**.
- **A fire that changes by itself:** make your own `i` from time or distance.
- **Your own building block** on Layer 3 (`Layers/OldExercises`).
- **A new room** with Steam Audio materials.

When I call everyone back

Play one fire. Tell us: the parts you heard; which worked better recorded or procedural; **one number you changed and what it did**. Then **format the SD card** and hand the recorder back. **Monday, if you can:** bring one exhibit sound that **could** be dynamic.